Syllabus Template **FF No.: 654-A**

Course Code: Programming for Engineers

Teaching Scheme	Examination Scheme
Credits : 4	LAB: 20Marks
Lectures : 2 Hrs/week	MSE: 20 Marks
Practical : 4 Hrs/week	ESE :50 Marks
Tutorial : 0 Hrs/week	CP:10 Marks

Prerequisites:	
●	1. Basic programming 2. Mathematics
Course Objectives:	
●	Understand basics of Python Programming
●	Describe Python operators, functions, and string manipulation techniques.
●	Discuss decision-making and looping constructs in Python Programming.
●	Apply Python's built-in data structures, type conversion, built-in methods, and comprehensions to organize and manipulate data efficiently.
●	Develop modular Python programs using user-defined functions, function arguments, recursion, and parameter-passing techniques.
●	Create data visualizations and perform numerical computations using NumPy arrays and Matplotlib.
Course Outcomes:	
	After completion of the course, student will be able to
1.	Demonstrate Python programming fundamentals by writing and executing simple programs.
2.	Develop Python programs using operators, mathematical functions, and string manipulation techniques.
3.	Construct Python programs using decision-making and looping constructs.
4.	Implement Python programs using appropriate data structures, mutability concepts, built-in methods, and comprehensions to solve computational problems effectively.
5.	Construct Python programs using user-defined functions, lambda expressions, recursion, and appropriate parameter-passing techniques to solve computational problems.
6.	Demonstrate Python programs using NumPy for numerical computation and Matplotlib for effective data visualization.

Section1:

Unit I: Fundamentals of Python Programming(4 Hours)
Introduction of Program, different types of Programming language: low level language, assembly level language, high level programming language, scripting language. Features of Python, How to Run Python code, Identifiers, Reserved Keywords, Variables, Comments in Python, Indentation in Python, Input, Output statement.
Self-Learning Topics: Python Installation and Environment Setup Debugging Basics.

Unit II: Operators (5 Hours) Operators(5 Hours)
Operators: Arithmetic Operators, relational Operators, Logical Operators, bitwise operators, other stray operators, Operator Precedence, Mathematical Functions, Trigonometric Functions, Random Number Functions, Strings in python: subscript operator, indexing, slicing a string, string methods.
Self-Learning Topics: Regular Expressions (Regex), String Encoding and Unicode.

Unit III: Flow Control and Loops (5 Hours)
Decision Making: if statement, if…else statement, if..elif ..else statement, nested if statement; Loops: while loop, for loop, range () function, continue and break statement, while and for loop with else statement, pass statement.
Self-Learning Topics: Time Complexity of Loops, Enumerate and Zip Functions.

Section2:

Unit IV: In-built Data Structures in Python (4 Hours)
List, Tuple, Set, Dictionary; Mutable and Immutable Objects, Data Type Conversion, Built-in methods and comprehensions.
Self-Learning Topics: Searching and Sorting Techniques using Python Data Structures.

Unit V: Functions(5 Hours)
Function Definition, Function Calling, Function Arguments, Anonymous Functions (Lambda Functions), recursive function, Function with more than one return value, default value to parameter, keyword parameter.
Self-Learning Topics: Nested Functions, Doc strings and documentation of Python Functions.

Unit VI: File Handling, NumPy and Matplotlib(5 Hours)
Opening a File, Modes for Opening a File, Attributes of file object, closing a File, reading from a File, writing to a File, File Methods. Introduction to NumPy, Arrays, Array indexing, comparison of Array and List data types, Array math, Broadcasting. Matplotlib -Plot, subplots and images.
Self-Learning Topics: Exception Handling in File Operations, Basic Image Processing, Advanced Plotting Techniques using Seaborn.

Text Books: *(As per IEEE format)*	
1	"Programming And Problem Solving with Python", Ashok Kamthane and Amit Ashok Kamthane , ISBN 9789387067578
2	"Beginning Programming with Python for Dummies", John Paul Mueller , Wiley Publishing, ISBN 9788126553488
Reference Books: *(As per IEEE format)*	
1	DiveintoPython3:MarkPilgrim,JesseNoller, Wiley Publishing,ISBN9788184899115.
2	Python in Easy Steps, Tata McGraw Hill Education, ISBN 9789351343080
3	Learning Python, 5th Edition, Mark Lutz, O′Reilly, ISBN 978-1449355739

List of Practical's:

Assignments on Fundamentals of Python

1. Program to Calculate the Area of a circle and triangle.
2. Program to Swap Two Variables.
3. Program to Generate a Random Number.
4. Program to Convert Kilometers to Miles.
5. Program to find maximum of two numbers.
6. Program to check if a number is even or odd.
7. Program to check if a number is positive, negative or 0.
8. A school decided to replace the desks in three classrooms. Each desk sits two students. Given the number of students in each class, print the smallest possible number of desks that can be purchased. The program should read three integers: the number of students in each of the three classes, a, b and c respectively. In the first test there are three groups. The first group has 20 students and thus needs 10 desks. The second group has 21 students, so they can get by with no fewer than 11 desks. 11 desks is also enough for the third group of 22 students. So, we need 32 desks in total.
9. H hours, M minutes and S seconds are passed since the midnight ($0 \le H < 12$, $0 \le M < 60$, $0 \le S < 60$). Determine the angle (in degrees) of the hour hand on the clock face right now.
10. Given integer coordinates of three vertices of a rectangle whose sides are parallel to the coordinate axes, find the coordinates of the fourth vertex of the rectangle. In the first test the three given vertices are (1, 4), (1, 6), (7, 4). The fourth vertex is thus (7, 6).
11. There was a set of cards with numbers from 1 to N. One of the cards is now lost. Determine the number on that lost card given the numbers for the remaining cards. Given a number N, followed by $N - 1$ integers - representing the numbers on the remaining cards (distinct integers in the range from 1 to N). Find and print the number on the lost card.

Assignments on Operators

12. Program to find maximum of three numbers.
13. Program to check if a year is leap.
14. Program to check if a date is valid.
15. Program to find the roots of a quadratic equation.
16. Given a string. Delete from it all the characters whose indices are divisible by 3.
17. Given a sequence of integer numbers ending with the number 0. Determine the length of the widest fragment where all the elements are equal to each other.
18. In bowling, the player starts with 10 pins at the far end of a lane. The object is to knock all the pins down. For this exercise, the number of pins and balls will vary. Given the number of pins N and then the number of balls K to be rolled, followed by K pairs of numbers (one for each ball rolled), determine which pins remain standing after all the balls have been rolled. The balls are numbered from 1 to N (inclusive) for this situation. The subsequent number pairs, one for each K represent the start to stop (inclusive) positions of the pins that were knocked down with each role. Print a sequence of N characters, where "I" represents a pin left standing and "." represents a pin knocked down.
19. A timestamp is three numbers: a number of hours, minutes and seconds. Given two timestamps, calculate how many seconds is between them. The moment of the first timestamp occurred before the moment of the second timestamp.

Assignments on Flow Control and Loops

20. Program to display first n numbers.
21. Program to calculate factorial of a numbers.
22. Program to display numbers in the reverse order.
23. Program to check if a number is prime.
24. Program to calculate sum and average of first n numbers.
25. Program to display first n multiples of a number.
26. Program to display first n Fibonacci numbers.
27. Program to find the sum of digits of a number.

Assignments on In-built Data Structures

28. Program to Create and view elements of a list.
29. Program to Create and view elements of a tuple.
30. Program to access List Index and Values
31. Program to add two Lists.
32. Program to check if a List is Empty or Not.
33. Program to Find the Largest Number in a List.
34. Program to Find the Second Largest Number in a List.
35. Program to Put Even and Odd elements in a List into Two Different Lists.
36. Program to Find all Numbers in a Range which are Perfect Squares and Sum ofall Digits in the Number is Less than 10.

37. Program to Generate Random Numbers from 1 to 20 and Append Them to the List.
38. Program to Remove the Duplicate Items from a List.
39. Program to create and view a dictionary.
40. Program to create and view elements of a set.
41. Program to Print values of dictionary.
42. Program to print all keys of dictionary.
43. Program to insert and delete from dictionary.
44. Program to sort (ascending and descending) a dictionary by value.
45. Program to concatenate dictionaries to create a new one.
46. Program to check whether a given key already exists in a dictionary.
47. Program to merge two Python dictionaries.
48. Program to get the maximum and minimum value in a dictionary.
49. Program to Add a list of elements to a set.
50. Program to Update the first set with items that don't exist in the second set.
51. Program to Return a set of elements present in Set A or B, but not both.
52. Program to check if two sets have any elements in common.
53. Program to Remove items from set1 that are not common to both set1 and set2.

Assignments on Functions

54. Function to check if a number is even or odd.
55. Function to find the maximum of two numbers.
56. Function with keyword arguments
57. Function with default arguments.
58. Function to Find the Factorial of a Number Using Recursion.
59. Function to Find the Sum of the Digits of the Number Recursively.
60. Create a function named count vowels that accepts a string and returns the number of vowels in the string.
61. Write a function greet user that takes a user's name as a parameter and prints a personalized greeting message.
62. Given a sequence of integers that end with a 00. Print the sequence in reverse order. Don't use lists or other data structures. Use the force of recursion instead.

Assignments on File Handling

63. To keep record of students' data, manipulate files to store, update, and delete students' information.
64. Write a Python program that reads the content of a text file and prints it to the console.
65. Create a Python program that takes user input (a string) and writes it to a new text file.
66. Write a Python program that reads a text file, counts the number of words in it, and writes the word count to a new file.
67. Write a Python program that reads the content of one text file and writes it to another file.
68. Develop a Python program that counts the number of lines in a text file and displays the count.

69. Write a Python program that appends a user-provided string to the end of an existing text file.
70. To keep record of patients' medical data, manipulate files to store, update, and delete such information.
71. Write a python program to create two 3X3 random matrixes and perform following operation: (a) Addition (b) subtraction (c) multiplication and display shape, dimensions, dtype, Rank and flatten output of every o/p matrix.
72. Write a Python program to plot line chat, bar chart, pi chart, scatter chart, histogram for taking two different arrays as input.

Assignments on Matplotlib

73. A coaching center recorded the number of hours studied and marks scored by 7 students. Visualize this data in a way that clearly reveals the relationship between study hours and marks, ensuring the graph is easy to interpret at a glance.
74. The performance of three students (A, B, C) across 5 tests needs to be compared on a single visual. Design a chart that allows a viewer to immediately identify who is performing best and how each student's trend differs, without any confusion between the three data series.
75. You are given a student's marks in 5 subjects. Determine the most suitable chart type(s) to (a) compare performance across subjects, (b) show each subject's proportion of the total marks, and (c) illustrate how the values are distributed. Create the corresponding visualizations.
76. Your report must present four different perspectives of data (a trend, a comparison, a relationship, and a distribution) on a single page without any overlap or visual clutter. Design a layout that accommodates all four clearly and legibly.
77. Among your three charts, one is more significant than the others and requires more visual emphasis, while the remaining two serve as supporting visuals. Design a single figure layout that makes this hierarchy visually evident.
78. You are provided with an image file. Load it into Python, display it visually, and determine the underlying numerical structure of the image data — including what each dimension of that data represents.
79. A colleague claims that a color image and its black-and-white counterpart "contain the same information, just displayed differently." Design a visual comparison that could support or challenge this claim, including a way to interpret the intensity values shown in the grayscale version.
80. Given a photograph, demonstrate three different ways the underlying image data can be transformed using array-based operations alone (without any built-in image-editing function). Present the original image alongside your results so that each transformation is clearly visible.
81. Your visualization needs to be submitted in a report that accepts only image files, and a separate copy is required for a printable document. Determine the appropriate file formats to save your chart in for each use case.

82. Select a real-world dataset relevant to your field of study (e.g., sensor readings, academic marks, weather data) and design a single dashboard-style figure using at least three different chart types that best represents the story the data tells.
83. Demonstrate how the same dataset's graph appears under three different visual themes, presented side-by-side, so that a viewer could select the most suitable style for a presentation.
84. In a dataset with fluctuating values, a viewer should be able to identify the highest and lowest points immediately, without scanning the entire graph. Devise a way to highlight these specific points directly on the chart.
85. You are given a grid of numeric values (e.g., a 5×5 matrix). Create a visualization that allows someone to quickly identify which cells contain high versus low values, without reading each number individually.
86. A specific region of your chart or image must be visually flagged as "important" or "noteworthy" for a reviewer, without altering the underlying data. Determine a way to highlight that region directly on the visualization.
87. Design a report figure in which one primary chart dominates the layout while two supporting charts accompany it — ensuring the viewer immediately understands that all three belong to the same analysis, through consistent labelling.

Assignments on NumPy and Matplotlib

88. A college wants to analyze the marks of 100 students in five subjects using NumPy. Develop a Python program to store the marks in a NumPy array and generate useful academic insights.
89. An IT company wants to analyze employee salary data using NumPy to support HR decisions.
90. A hospital has collected patient health information and wants to identify patients requiring immediate medical attention.
91. A weather department records daily temperatures for one month. Analyze the temperature data using NumPy.
92. A smart city project records hourly vehicle counts at a traffic signal. Analyze the traffic pattern using NumPy.
93. An online shopping company wants to analyze monthly sales of five products.
94. A manufacturing company measures the dimensions of produced components. Components outside the acceptable range are considered defective.
95. Images are represented as NumPy arrays. Develop a program to perform basic image enhancement operations.
96. A fitness application records daily step counts for users. Analyze their activity using NumPy.
97. A cricket coach wants to analyze a player’s performance over 20 matches using NumPy.
98. Stress–Strain Analysis: Calculate stress and strain from experimental data using NumPy and plot the stress–strain curve using Matplotlib
99. Temperature Distribution in a Metal Rod: Analyze temperature measurements at different positions along a rod and visualize the temperature profile.

100. Spring–Mass System Simulation: Simulate the displacement of a spring–mass system using NumPy and visualize the motion over time.
101. Manufacturing Quality Control: Analyze dimensions of manufactured components and plot histograms to study dimensional variation.
102. Fuel Consumption Analysis: Compare fuel consumption of vehicles at different speeds and plot the results.
103. Traffic Volume Analysis: Analyze hourly traffic count data and plot traffic flow patterns.
104. Rainfall Data Analysis: Process monthly or annual rainfall data and visualize rainfall trends using line and bar charts.
105. Load–Deflection Analysis of a Beam: Calculate beam deflection under different loads and plot load versus deflection.
106. Concrete Compressive Strength Analysis: Analyze compressive strength test results and visualize the variation among samples.
107. Bridge Vibration Monitoring: Process vibration sensor data from a bridge and plot vibration trends over time.
108. Water Consumption Analysis: Analyze daily water consumption data for a residential area and visualize usage patterns.
109. Construction Project Cost Analysis: Compare planned and actual project costs and display the results using charts.
110. Soil Property Analysis: Analyze soil moisture or density data collected from different locations and visualize the distribution.

List of Project areas:

1. Web Application
2. Data analysis
3. Computer vision
4. Machine learning
5. Artificial intelligence
6. Web technologies
7. Numerical analysis
8. Robotics & Automation
9. Manufacturing
10. Simulation & Modeling
11. Graph Plotting & Visualization
12. Quality Control
13. Construction Management
14. Building Information Modeling (BIM)
15. Geotechnical Engineering
16. Hydrology & Water Resources
17. Surveying & GIS
18. Structural Analysis
19. IoT
20. Smart Infrastructure

CO – PO / PSO Articulation Matrix

Correlation: 1 = Low (Slight), 2 = Medium (Moderate), 3 = High (Substantial), '-' = no correlation.

CO	Program Outcome (PO)											PSO		
CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO 1	PSO 2	PSO 3
CO1	3	2	-	-	2	-	-	-	-	-	-			
CO2	3	3	2	-	2	-	-	-	-	-	-			
CO3	3	3	2	1	2	-	-	-	1	-	-			
CO4	3	3	3	2	3	-	-	-	1	-	-			
CO5	3	3	3	2	3	-	-	-	2	1	-			
CO6	2	3	2	2	3	-	-	-	1	1	-			
Avg	2.83	2.83	2.40	1.75	2.50	-	-	-	1.25	1.00	-			

CO – WK Mapping (Knowledge & Attitude Profiles) Sample Table

'✓' indicates the knowledge profile drawn upon by the CO.

CO	WK1	WK2	WK3	WK4	WK5	WK6	WK7	WK8	WK9
CO1									
CO2									
CO3									
CO4									
CO5									
CO6									

CO – WP Mapping (Complex Engineering Problems) Sample Table

CO	WP1	WP2	WP3	WP4	WP5	WP6	WP7
CO1							
CO2							
CO3							
CO4							
CO5							
CO6							

CO – EA Mapping (Complex Engineering Activities)

'✓' indicates the activity attribute addressed by the CO

CO	EA1	EA2	EA3	EA4	EA5
CO1					
CO2					
CO3					
CO4					
CO5					
CO6					

CO – SDG Mapping (Sustainable Development Goals) Sample Table

CO	SDG(s) Addressed	Justification / Linkage
CO1		
CO2		
CO3		
CO4		
CO5		
CO6		

